Ex. No: 11b
Date: 13/9/26

Simulate RIP using CISCO Packet Tracer

AIM:

To simulate and configure RIP (routing information Protocol) using CISCO Packet Tracer and verify communication between different networks.

PROCEDURE:

1. Create the network
   * Place 3 routers (Router 0, Router 1, Router 2) and 2 PC's (PC0, PC1)
   * Connect the routers using serial links and PC's through Fast Ethernet.
   * Assign the given IP addresses to all PC's and router interfaces.
2. Configure Router interfaces.
   * Open router → CLI and enter: enable, configure terminal, interface < interface name>
   * For serial interfaces, configure clock rate 64000 and bandwidth 64 where required.
3. Configure RIP
   * Enable RIP on each routes.
   * Advertize all directly connected networks
   * Repeat the RIP configuration on Router 1 and Router 2 with their repective connected networks.
4. Verify Connectivity.
   * Open PC → Desktop → command prompt
   * Test communication
   * Verify that PC0 can successfully communicate with PC1 through the routers.

RIP Routing :-

```
Router > enable
Router # configure terminal
Router (config) # router rip
Router (config-router) # version 2
Router (config-router) # network 192.168.10.
Router (config-router) # network 192.168.2.0
Router (config-router) # end
Router # show ip route
```

Sample output :-

R 192.168.2.0 /24 [120/1] via 10.0.0.2

8/8

5. Verify Dynamic routing

* Use tracert to check the route taken by packets
* Disconnect one routing link and test again.
* RIP automatically selects an alternative available route if the current route fails.

8e

RESULT:-

The simulation of RIP using CISCO Packet Tracer has been implemented successfully.